

Motivation: topology in clustering

Classical clusterings (k -means, GMM, DBSCAN, spectral) only see **connected components** (H_0). Real point clouds carry higher-order topology: **loops** (H_1 , rings/holes) and **voids** (H_2 , cavities – e.g. pockets in molecules).

Goal. Cluster points using loops/voids *and* the way these topological features *intersect*.

Vietoris–Rips complex & Hodge Laplacian

From a point cloud $X \subset \mathbb{R}^d$ build a Vietoris–Rips complex $\mathcal{S}$ of radius ε . With boundary operators B_0, B_1, B_2 , the Hodge Laplacians are

$$L_k = B_{k-1}^\top B_{k-1} + B_k B_k^\top.$$

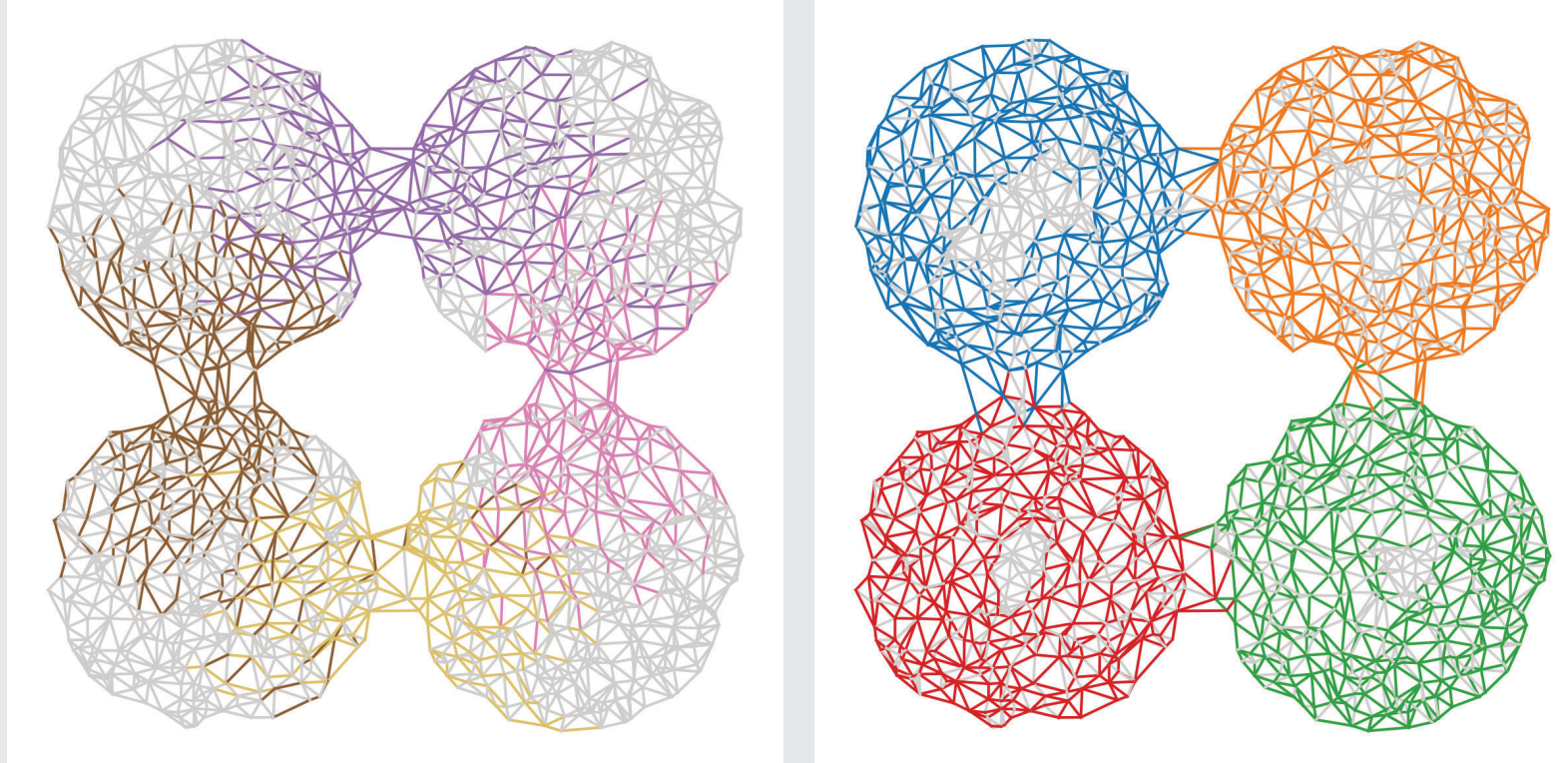
Hodge theorem: $\dim \ker L_k = \beta_k$ (k -th Betti number) – the kernel encodes the k -dimensional homology.

Hodge decomposition: the key tool

For every dimension k the chain space splits orthogonally

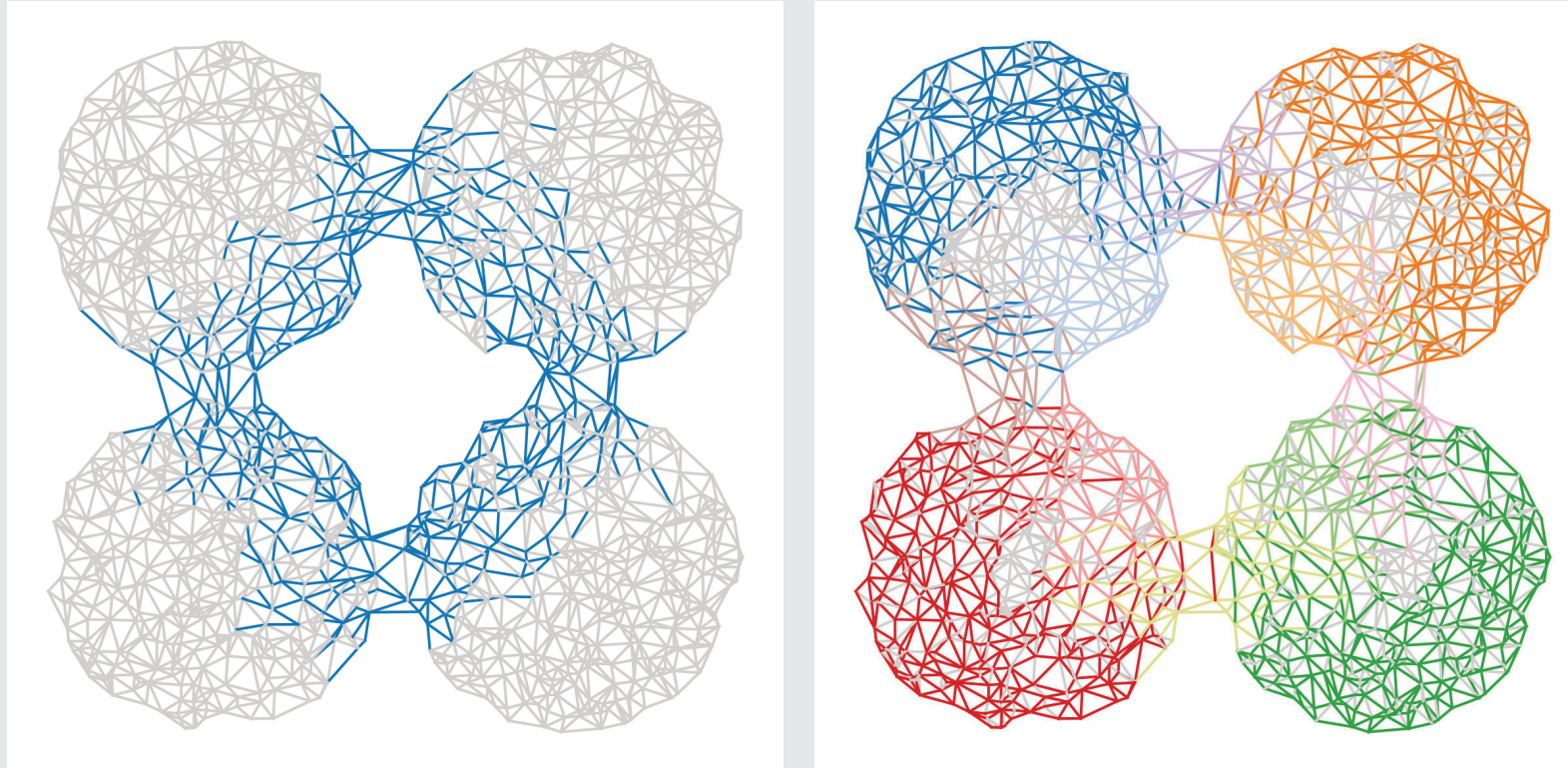
$$\mathbb{R}[\mathcal{S}_k] = \underbrace{\text{Im} B_{k-1}^\top}_{\text{gradient}} \oplus \underbrace{\ker L_k}_{\text{harmonic}} \oplus \underbrace{\text{Im} B_k}_{\text{curl}}$$

Only the **harmonic** part encodes topology: each basis vector of $\ker L_k$ is a representative of a non-trivial homology class (loop for $k = 1$, void for $k = 2$).



gradient

curl



harmonic

full signal

Clustering edges by the three Hodge components: only the harmonic part separates the two loops [1].

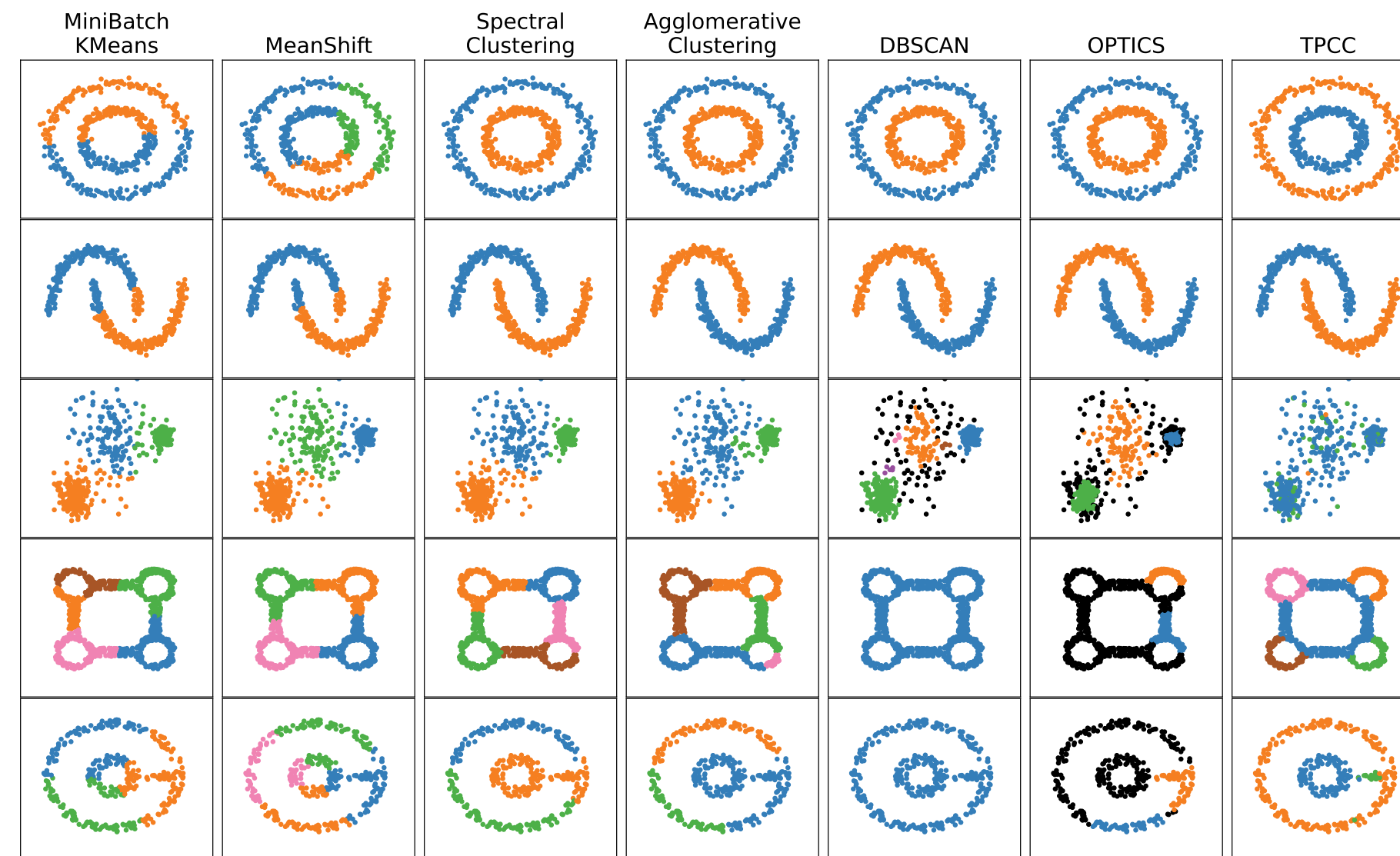
Explicit projections on edges ($k = 1$):

$$c_1^G = B_0^\top (B_0 B_0^\top)^{-1} B_0 c_1, \quad c_1^S = B_1 (B_1^\top B_1)^{-1} B_1^\top c_1, \\ c_1^H = c_1 - c_1^G - c_1^S.$$

TPCC – Topological Point Cloud Clustering [3]

Algorithm 1 Topological Point Cloud Clustering (TPCC)

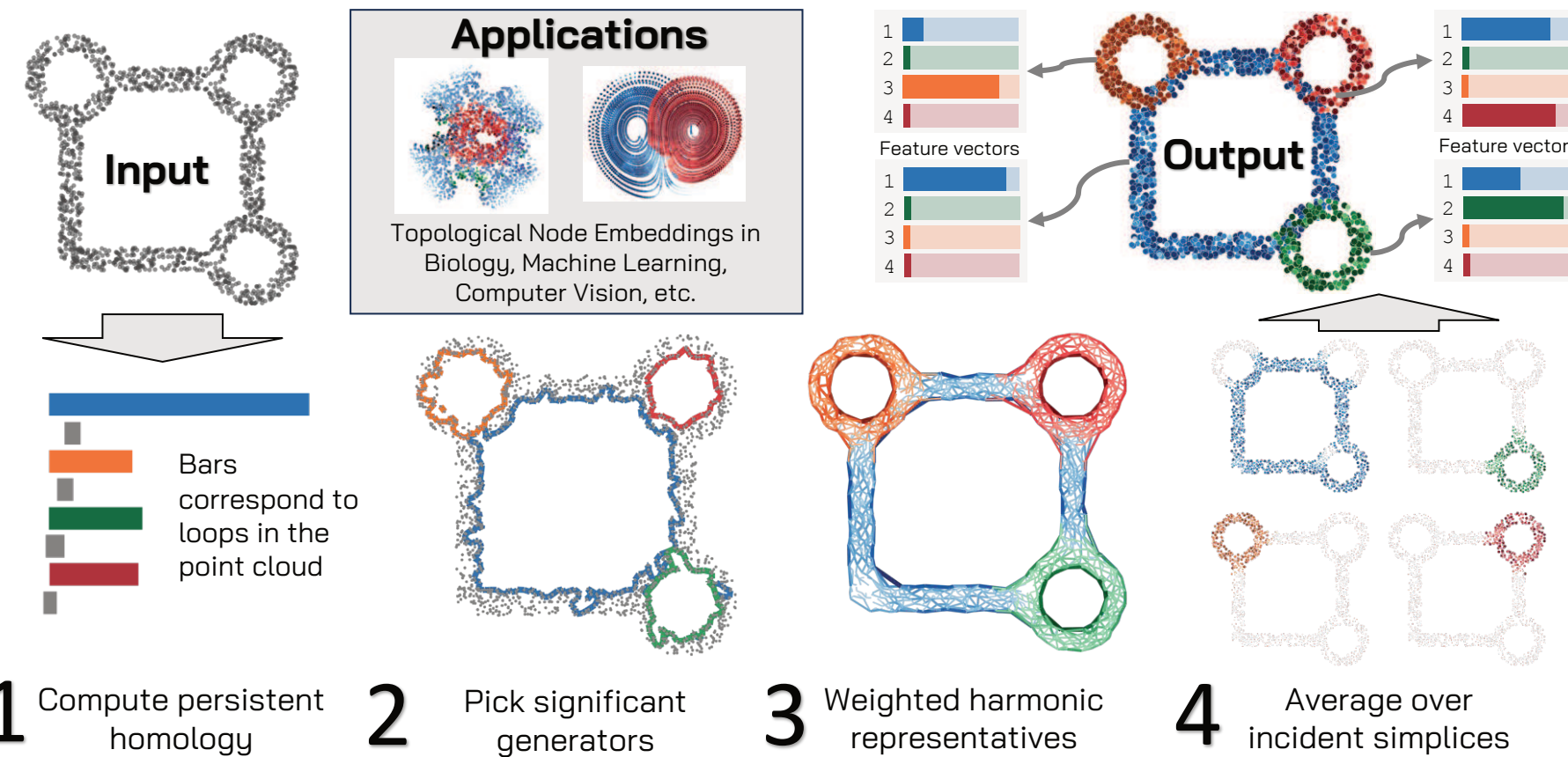
- 1: **Input:** point cloud X , max. dimension d
- 2: Pick ε , build Vietoris–Rips complex $\mathcal{S}$ of X
- 3: Build Hodge Laplacians $L_0, \dots, L_d$
- 4: **for** $i = 0$ to d **do**
- 5: Compute 0-eigenvectors $v_0^i, \dots, v_{\beta_i}^i$ of L_i
- 6: Sub-space cluster the rows of $[v_0^i, \dots, v_{\beta_i}^i]$
- 7: Assign cluster labels to i -simplices
- 8: **for** $x \in X$ **do** collect labels of i -simplices $\sigma \ni x$ (*signature of x*)
- 9: **end for**
- 10: **end for**
- 11: Cluster X on topological signatures
- 12: **Output:** cluster label for every $x \in X$



Classical clusterings vs. TPCC. Only TPCC separates points by which loop/void they belong to [3].

TOPF – Point-level Topological Features [2]

TOPF turns “which homology class does x contribute to” into a continuous **point-level** feature vector: persistence filtering $\rightarrow$ indicator χ for each stable class $\rightarrow$ **harmonic projection** on $\ker L_k \rightarrow$ aggregation to vertices, giving $F \in \mathbb{R}^{N \times M}$.



TOPF pipeline: point cloud $\rightarrow$ Rips complex $\rightarrow$ persistent generator $\rightarrow$ harmonic projection $\rightarrow$ per-point feature vector [2].

This work: intersection graphs

TPCC/TOPF say *which* topological object each point belongs to, but not *how those objects relate*: are two loops linked? does a ring pierce a sphere?

Idea. Build an **intersection graph** $G = (V, E)$: vertices are harmonic representatives of H_1/H_2 classes; edges carry a similarity weight W_{uv} between features.

Cross-dimensional feature similarity

Loops live on edges, voids on triangles. Cosine of harmonic vectors (using B_1 to bridge dimensions) gives

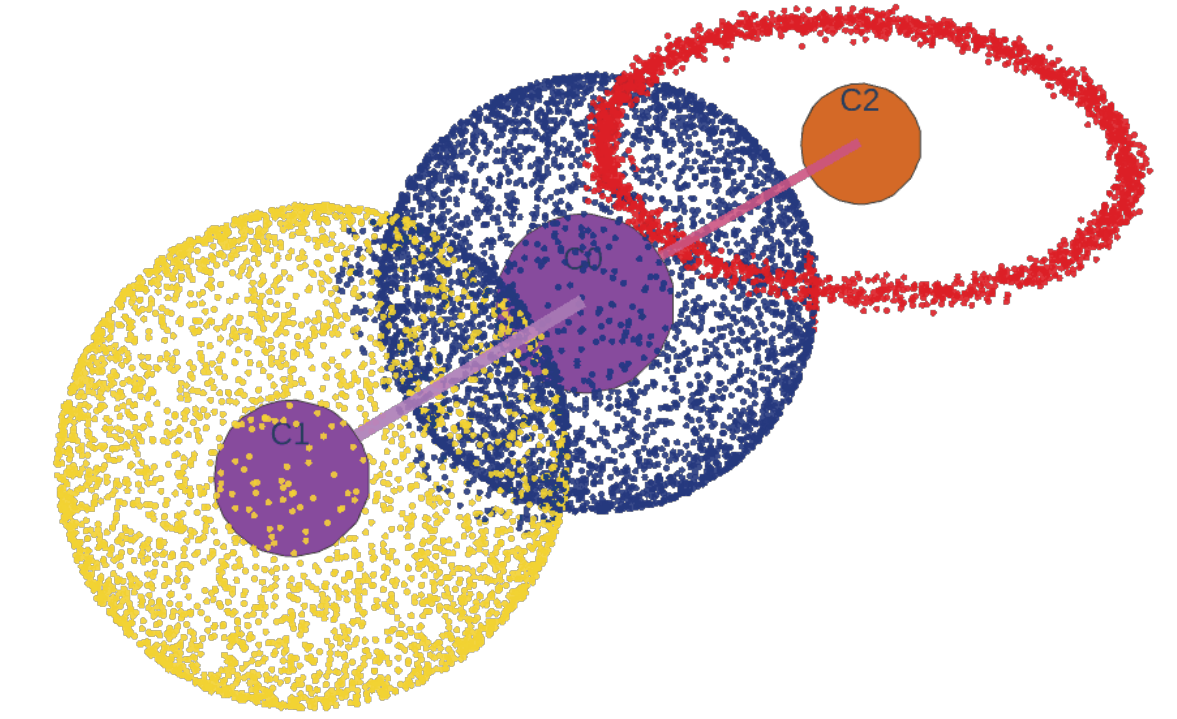
$$W_{ij}^{LL} = \cos(h_1^{(i)}, h_1^{(j)}), \quad W_{ij}^{LV} = \cos(h_1^{(i)}, B_1 h_2^{(j)}), \quad W_{ij}^{VV} = \cos(h_2^{(i)}, h_2^{(j)}).$$

For a *structurally aware* similarity, cohomological **cap/cup products** of harmonic cocycles α, β :

$$(\alpha \frown \beta)_v = \sum_{e=(u,v)} \alpha_e \beta_e, \quad (\alpha \smile \beta)_{(v_0 v_1 v_2)} = \alpha_{(v_0 v_1)} \beta_{(v_1 v_2)}.$$

Well-defined on cohomology classes – independent of representative.

Intersection graph on a synthetic cloud



“Two spheres + ring” in $\mathbb{R}^3$: two H_2 nodes, one H_1 node. The loop wires only to the sphere it pierces; edge colour = pair type.

Experiments: MNIST and CIFAR-10

AE $\rightarrow$ PCA $\rightarrow$ TOPF $\rightarrow$ optional CAP/CUP (KNN-prop.) $\rightarrow$ KMeans / DEC / CC.

Clustering	TOPF	CAP	CUP	MNIST					CIFAR-10				
				ACC	NMI	RI	FMI	BCubed	ACC	NMI	RI	FMI	BCubed
KMeans	✓			0.79	0.78	0.71	0.75	0.76	0.25	0.13	0.07	0.17	0.17
	✓	✓		0.8	0.79	0.73	0.76	0.78	0.25	0.13	0.07	0.17	0.17
	✓		✓	0.86	0.79	0.75	0.78	0.78	0.25	0.13	0.07	0.17	0.17
DEC				0.87	0.87	0.83	0.84	0.87	0.2	0.07	0.04	0.2	0.19
				0.57	0.47	0.4	0.46	0.47	0.22	0.09	0.05	0.14	0.14
CC	✓			0.61	0.47	0.42	0.48	0.48	0.22	0.08	0.05	0.14	0.14
	✓		✓	0.66	0.5	0.47	0.52	0.52	0.22	0.09	0.04	0.14	0.14

References

- [1] Vincent P. Grande and Michael T. Schaub. Disentangling the spectral properties of the Hodge Laplacian: Not all small eigenvalues are equal. In *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2024.
- [2] Vincent P. Grande and Michael T. Schaub. Point-level topological representation learning on point clouds. In *42nd International Conference on Machine Learning (ICML)*, 2025.
- [3] Vincent P. Grande Grande and Michael T. Schaub. Topological point cloud clustering. In *40th International Conference on Machine Learning (ICML)*, 2023.